

بنك أسئلة الاقتصاد الهندسي

1. An interest rate of 10% per year compounded semiannually is the same as:

a. an effective 10% per month.

b. an effective 10.25% per year. ✓

c. an effective 5% per quarter compounded continuously.

2. An interest rate of 8% per year compounded monthly is the same as:

a. an effective 8.29% per year. ✓

b. an effective 8.29% per month.

c. an effective 8% per year.

3. An interest rate of 8% per year compounded yearly is the same as:

a. an effective 8.29% per year

b. an effective 8% per year ✓

c. an effective 8.29% per month

4. An interest rate of 12% per semiannually compounded monthly is the same as:

a. an effective 26.8% per year.

b. an effective 1.5% per month.

c. an effective 12.16% per year. ✓

5. An interest rate of 1.5% per month compounded continuously is the same as:

a. an effective 1.511% per month.

b. an effective 1.5% per month.

c. an effective 18% per year. ✓

6. An interest rate of 5% per month compounded continuously is the same as:

a. an effective 5% per month.

b. an effective 2.5% per quarter.

c. an effective 5.12% per month. ✓

is: $(A/G, i, n)(P/A, i, n) = (----?----)$ 7. The missing factor in the equation

$(G/P, i, n)$ a.

$(P/G, i, n)$ b. ✓

$(A/P, i, n)$ c.

is: $(A/G, i, n)(P/A, i, n) = (P/G, i, n)$ 8. The missing factor in the equation

$(G/P, i, n)$ a.

$(P/G, i, n)$ b. ✓

$(A/P, i, n)$ c.

is: $(A/P, i, n)(P/A, i, n) = (----?----)$ 9. The missing factor in the equation

$(P/G, i, n)$ a.

$(G/P, i, n)$ b.

$(A/P, i, n)$ c. ✓

is: $(----?----)(A/P, i, n) = (A/F, i, n)$ 10. The missing factor in the equation

$(P/A, i, n)$ a.

$(F/P, i, n)$ b.

$(P/F, i, n)$ c. ✓

is: $(P/F, i, n)(----?----) = (A/F, i, n)$ 11. The missing factor in the equation

$(P/A, i, n)$ a.

$(F/P, i, n)$ b.

$(A/P, i, n)$ c. ✓

is: $(P/A, i, n)(F/P, i, n) = (----?----)$ 12. The missing factor in the equation

$(P/F, i, n)$ a.

$(F/A, i, n)$ b. ✓

$(A/F, i, n)$ c.

is: $(P/A, i, n)(F/P, i, n) = (F/A, i, n)$ 13. The missing factor in the equation

$(F/A, i, n)$ a.

$(A/F, i, n)$ b.

$(P/F, i, n)$ c. ✓

is: $(----?----)(F/P, i, n) = (F/A, i, n)$ 14. The missing factor in the equation

$(P/A, i, n)$ a. ✓

$(F/A, i, n)$ b.

$(F/P, i, n)$ c.

is: $(F/A, i, n)(----?----) = (P/A, i, n)$ 15. The missing factor in the equation

$(F/A, i, n)$ a.

$(A/F, i, n)$ b.

$(P/F, i, n)$ c. ✓

peri = 10%16. For an original (first) amount of JD 1500, at a Simple Interest Rate year, the Interest Amount (I) after 2 years is:

a. JD 1800

b. JD 300✓

c. JD 30,000

peri = 10%17. For an original (first) amount of JD 1500, at a Simple Interest Rate year, the Withdrawal (Total) Amount after 2 years is:

a. JD 300

b. JD 30,000

c. JD 1800✓

peri = 10%18. For an original (first) amount of JD 1500, at a Simple Interest Rate year, the Interest Amount (I) after 3 years is:

a. JD 1950

b. JD 450✓

c. JD 45,000

peri = 10%19. For an original (first) amount of JD 1500, at a Simple Interest Rate year, the Interest Amount (I) after 5 years is:

a. JD 2250

b. JD 75,000

c. JD 750✓

peri = 10%20. For an original (first) amount of JD 1500, at a Simple Interest Rate year, the Interest Amount (I) after 7 years is:

a. JD 1050✓

b. JD 2550

c. JD 105,000

peri = 10%21. For an original (first) amount of JD 1500, at a Simple Interest Rate year, the Withdrawal (Total) Amount after 7 years is:

a. JD 105,000

b. JD 2550✓

c. JD 1050

peri = 10%22. For an original (first) amount of JD 1500, at a Simple Interest Rate year, the Interest Amount (I) after 9 years is:

a. JD 135,000

b. JD 1350 ✓

c. JD 2850

peri = 10%23. For an original (first) amount of JD 1500, at a Simple Interest Rate year, the Interest Amount (I) after 11 years is:

a. JD 3150

b. JD 165,000

c. JD 1650 ✓

peri = 10%24. For an original (first) amount of JD 1500, at a Simple Interest Rate year, the Interest Amount (I) after 12 years is:

a. JD 3300

b. JD 180,000

c. JD 1800 ✓

25. A project with an annual operating cost = JD 1000, increased by 100 yearly over per year, then the equivalent single payment at time zero (Eq. $i = 8\%$ years. If $n = 4$ $X(0)$) is around:

a. 3777.1✓

b. 244.5

c. 2821.6

26. A project with an annual operating cost = JD 1000, increased by 100 yearly over per year, then the equivalent single payment at time zero (Eq. $i = 8\%$ years. If $n = 6$ $X(0)$) is around:

a. 4729.94

b. 5675.1

c. 1052.3✓

27. A project with an annual operating cost = JD 1000, increased by 100 yearly over per year, then the equivalent single payment at time zero (Eq. $i = 8\%$ years. If $n = 7$ $X(0)$) is around:

a. 1402.4

b. 5675.1

c. 6608.7✓

28. A project with an annual operating cost = JD 1000, increased by 100 yearly over per year, then the equivalent single payment at time zero (Eq. $i = 8\%$ years. If $n = 8$ $X(0)$) is around:

a. 6608.7

b. 7527.2✓

c. 1780.6

29. A project with an annual operating cost = JD 1000, increased by 100 yearly over per year, then the equivalent single payment at time zero (Eq. $i = 8\%$ years. If $n = 9$ $X(0)$) is around:

a. 8427.6✓

b. 7527.2

c. 2180.8

30. A project with an annual operating cost = JD 1000, increased by 100 yearly over per year, then the equivalent single payment at time zero (Eq. $i = 8\%$ years. If $n = 11$ $X(0)$) is around:

a. 9307.6

b. 3026.5

c. 10165.4✓

31. A project with an annual operating cost = JD 1000, increased by 100 yearly over per year, then the equivalent single payment at time zero (Eq. $i = 8\%$ years. If $n = 12$ $X(0)$ is around:

a. 10499.3 ✓

b. 3463.3

c. 10999.3

32. A project with an annual operating cost = JD 1000, increased by 100 yearly over per year, then the equivalent single payment at time zero (Eq. $i = 8\%$ years. If $n = 13$ $X(0)$ is around:

a. 3904.6

b. 10999.3

c. 11808.3 ✓

per year $i = 8\%$ 33. A project with an Annual Income for the first 4 years = 1000, if years, then the Equivalent Annual Value for the last 4 years is closest to: $n = 12$ and

a. -1851

b. -540.3 ✓

c. -1000

years. Then, the Eq. $n = 11$ per year and $i = 8\%$ ³⁴. Given a Salvage Value = JD 1000;
X(1) (equivalent single value at the end of year 1) is:

a. -463.2 ✓

b. -1080

c. -428.88

years. Then, the Eq. $n = 11$ per year and $i = 8\%$ ³⁵. Given a Salvage Value = JD 1000;
X(2) (equivalent single value at the end of year 2) is:

a. -1999

b. -428.88

c. -500.25 ✓

years. Then, the Eq. $n = 11$ per year and $i = 8\%$ ³⁶. Given a Salvage Value = JD 1000;
X(3) (equivalent single value at the end of year 3) is:

a. -1850.9

b. -540.27 ✓

c. -428.88

years. Then, the Eq. $n = 11$ per year and $i = 8\%$ ³⁷. Given a Salvage Value = JD 1000;
X(4) (equivalent single value at the end of year 4) is:

a. -428.88

b. -1713.3

c. -583.49 ✓

years. Then, the Eq. $n = 11$ per year and $i = 8\%$ ³⁸. Given a Salvage Value = JD 1000;
X(5) (equivalent single value at the end of year 5) is:

a. -630.17 ✓

b. -428.88

c. -11586.8

years. Then, the Eq. $n = 11$ per year and $i = 8\%$ ³⁹. Given a Salvage Value = JD 1000;
X(7) (equivalent single value at the end of year 7) is:

a. -1360.4

b. -428.88

c. -735.03 ✓

years. Then, the Eq. $n = 11$ per year and $i = 8\%$ 40. Given a Salvage Value = JD 1000;
X(8) (equivalent single value at the end of year 8) is:

a. -428.88

b. -793.83✓

c. -1259.7

years. Then, the Eq. $n = 11$ per year and $i = 8\%$ 41. Given a Salvage Value = JD 1000;
X(10) (equivalent single value at the end of year 10) is:

a. -428.88

b. -1080

c. -925.93✓

42. For a project with an Extra Cost at the end of the 8th year = 1300 and a S.V. =
years, then the Equivalent Single Value at the end $n = 12$ per year and $i = 8\%$ 500. If
of the 8th year X(8) is:

a. -932.5✓

b. +932.5

c. -1667.5

43. For a project with an Extra Cost at the end of the 9th year = 1300 and a S.V. = years, then the Equivalent Single Value at the end $n = 12$ per year and $i = 8\%$ 500. If of the 9th year $X(9)$ is:

a. +903.1

b. -1696.9

c. -903.1 ✓

44. Given a Project with the following data: First (initial) cost = JD 7000; Annual $n = 12$ per year, and $i = 8\%$ Operating cost = JD 1200; Salvage Value = JD 1500; years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 1, we use:

$7000(F/P, 8\%, 1) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 11)$ a.

$7000(F/P, 8\%, 1) + 1200(P/A, 8\%, 12)(F/P, 8\%, 1) - 1500(P/F, 8\%, 11)$ b.

$7000 + 1200(P/A, 8\%, 12)(F/P, 8\%, 1) - 1500(P/F, 8\%, 11)$ c. ✓

45. Given a Project with the following data: First (initial) cost = JD 7000; Annual $n = 12$ per year, and $i = 8\%$ Operating cost = JD 1200; Salvage Value = JD 1500; years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 4, we use:

$7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12)(F/P, 8\%, 4) - 1500(P/F, 8\%, 8)$ a. ✓

$7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 8)$ b.

$7000(F/P, 8\%, 3) + 1200(P/A, 8\%, 12)(F/P, 8\%, 3) - 1500(P/F, 8\%, 8)$ c.

46. Given a Project with the following data: First (initial) cost = JD 7000; Annual $n = 12$ per year, and $i = 8\%$ Operating cost = JD 1200; Salvage Value = JD 1500; years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 5, we use:

$$7000(F/P, 8\%, 5) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 7) \text{a.}$$

$$7000(F/P, 8\%, 5) + 1200(P/A, 8\%, 12)(F/P, 8\%, 5) - 1500(P/F, 8\%, 7) \text{b. } \checkmark$$

$$7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12)(F/P, 8\%, 4) - 1500(P/F, 8\%, 7) \text{c.}$$

47. Given a Project with the following data: First (initial) cost = JD 7000; Annual $n = 12$ per year, and $i = 8\%$ Operating cost = JD 1200; Salvage Value = JD 1500; years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 6, we use:

$$7000(F/P, 8\%, 6) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 6) \text{a.}$$

$$7000(F/P, 8\%, 5) + 1200(P/A, 8\%, 12)(F/P, 8\%, 5) - 1500(P/F, 8\%, 6) \text{b.}$$

$$7000(F/P, 8\%, 6) + 1200(P/A, 8\%, 12)(F/P, 8\%, 6) - 1500(P/F, 8\%, 6) \text{c. } \checkmark$$

48. Given a Project with the following data: First (initial) cost = JD 7000; Annual $n = 12$ per year, and $i = 8\%$ Operating cost = JD 1200; Salvage Value = JD 1500; years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 7, we use:

$$7000(F/P, 8\%, 7) + 1200(P/A, 8\%, 12)(F/P, 8\%, 7) - 1500(P/F, 8\%, 5) \text{a. } \checkmark$$

$$7000(F/P, 8\%, 7) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 5) \text{b.}$$

$$7000(F/P, 8\%, 6) + 1200(P/A, 8\%, 12)(F/P, 8\%, 6) - 1500(P/F, 8\%, 5) \text{c.}$$

49. Given a Project with the following data: First (initial) cost = JD 7000; Annual $n = 12$ per year, and $i = 8\%$ Operating cost = JD 1200; Salvage Value = JD 1500; years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 8, we use:

$$7000(F/P, 8\%, 8) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 4) \text{a.}$$

$$7000(F/P, 8\%, 7) + 1200(P/A, 8\%, 12)(F/P, 8\%, 7) - 1500(P/F, 8\%, 4) \text{b.}$$

$$7000(F/P, 8\%, 8) + 1200(P/A, 8\%, 12)(F/P, 8\%, 8) - 1500(P/F, 8\%, 4) \text{c. } \checkmark$$

50. Given a Project with the following data: First (initial) cost = JD 7000; Annual $n = 12$ per year, and $i = 8\%$ Operating cost = JD 1200; Salvage Value = JD 1500; years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 10, we use:

$$7000(F/P, 8\%, 9) + 1200(P/A, 8\%, 12)(F/P, 8\%, 9) - 1500(P/F, 8\%, 2) \text{a.}$$

$$7000(F/P, 8\%, 10) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 2) \text{b.}$$

$$7000(F/P, 8\%, 10) + 1200(P/A, 8\%, 12)(F/P, 8\%, 10) - 1500(P/F, 8\%, 2) \text{c. } \checkmark$$